

CATALOGUE

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1. Introduction

An imbalanced wheel will make the wheel jump and steering wheel wobble while driving. It can baffle the driver to drive, aggrandize the cleft of combine area of steering system, damage the vibration damper and steering parts, and increase the probability of the traffic accidents. A balanced wheel will avoid all these problems.

This equipment adopts the new LSI (Large Scale Integrated circuit) to constitute the hardware system that acquires processes and calculates information at a high speed.

Read the manual carefully before operating the equipment to ensure normal and safe operation. Dismantling or replacing the parts of equipment should be avoided. When it needs repairing, please contact with technique service department. Before balancing, ensure the wheel fixed on the flange tightly. Operator should wear close-fitting smock to prevent from hanging up. Non-operator does not start the equipment.

No use while beyond the stated function range of manual.

2. Specification and Features

2.1 Specification

Max wheel weight: 65kg

Motor power: 200W

Power supply: 220V/50Hz

Balancing precision: $\pm 1\text{g}$

Rotating speed: 200r/min

Cycle time: 8s

Rim diameter: 10"~24"(256mm~610mm)

Rim width: 1.5"~20"(40mm~510mm)

Noise: $< 70\text{dB}$

Net weight: 98kg

Dimensions: 915mm×760mm×1180mm

2.2 Features

Adopt 9 LED display, it has flexible indicator operating function.

Various balancing modes can carry out counterweights to stick, clamp, or hidden stick etc.

Input data of rim automatically by measure scale.

Intelligent self-calibrating and measure scale self-labeling function.

Self fault diagnosis and protection function.

Applicable for various rims of steel structure and duralumin structure.

2.3 Working Environment

Temperature: 5~50℃

Height above sea level: $\leq 4000\text{m}$

Humidity: $\leq 85\%$

3. The Constitution of Dynamic Balancer

Two major components of the dynamic balancer are: machine and electricity:

3.1 Machine

The part of machine consists of support, swing support and main axis; they are together fixed on the frame.

3.2 Electricity system

1. The microcomputer system is made up of the LSI such as new high speed MCU CPU system and keyboard.
2. Automatic measure scale.
3. Testing speed and positioning system consists of gear and opto-electronic

- coupler.
4. Two-phase asynchronous motor supplies and controlling circuit.
 5. Horizontal and vertical pressure sensor.
 6. Hood protection.

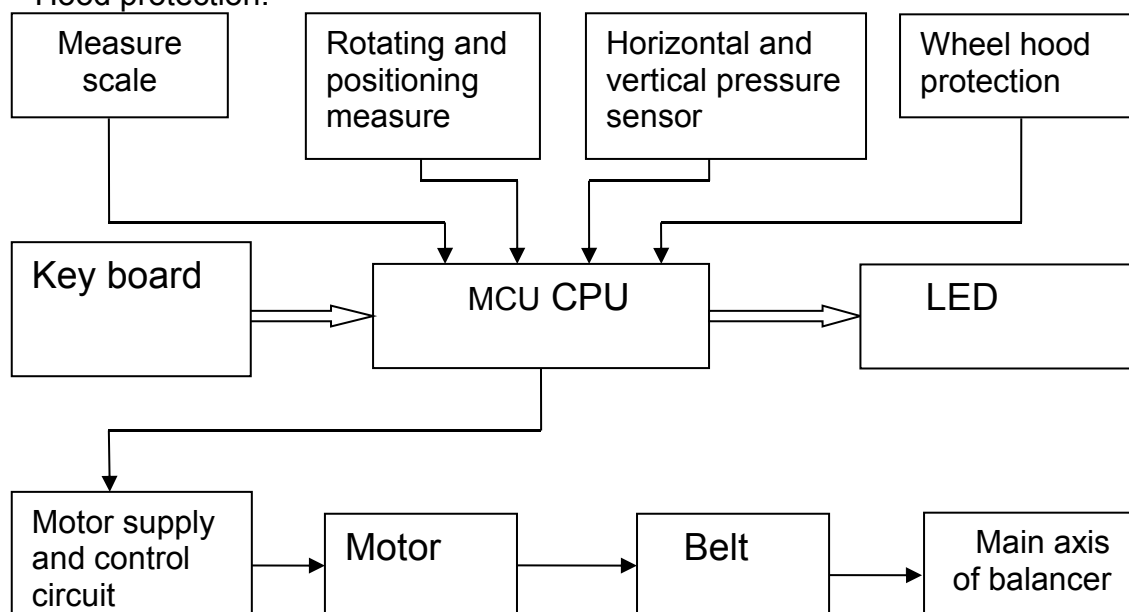


Figure 3-1

4. Installation of Dynamic Balancer

4.1 Opening and Checking

Open the package and check whether there are damaged parts. If there are some problems, please do not use the equipment and contact with the supplier. Standard accessories with equipment are shown as follow:

Screw stud of drive axis	1
Balancing pliers	1
Allen wrench	1
Measure caliper	1
Locking nut	1
Adapter (cone)	4
Counterweight (100g)	1
Protection hood (optional)	1

4.2 Installing machine

4.2.1 The balancer must be installed on the solid cement or similar ground, unsolidified ground can bring measuring errors.

4.2.2 There should be 50cm around the balancer in order to operate conveniently.

4.2.3 Nail anchor bolts on the base's mounting hole of balancer to fix the balancer.

4.3 Installing hood

Install the frame of hood on the equipment (optional): plug the pipe of protection hood into the hood shift (behind the cabinet), then fix with M10×65 screws.

4.4 Installing screw stud of drive axis

Install screw stud of drive shaft on the main shaft. (Refer to figure 4-1)

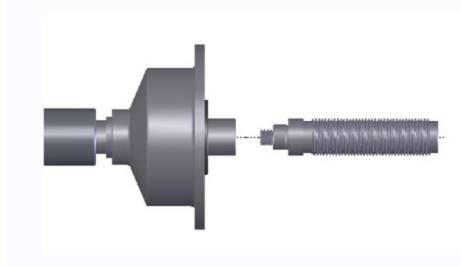


Figure 4-1

(**Notice:** a wheel can be installed on the main axis before screwing down, then hold the wheel by hands in order to prevent the main axis revolving together with the bolt.)

5. LED display control panel and function keys

Figure 5-1 displaying on the LED panel

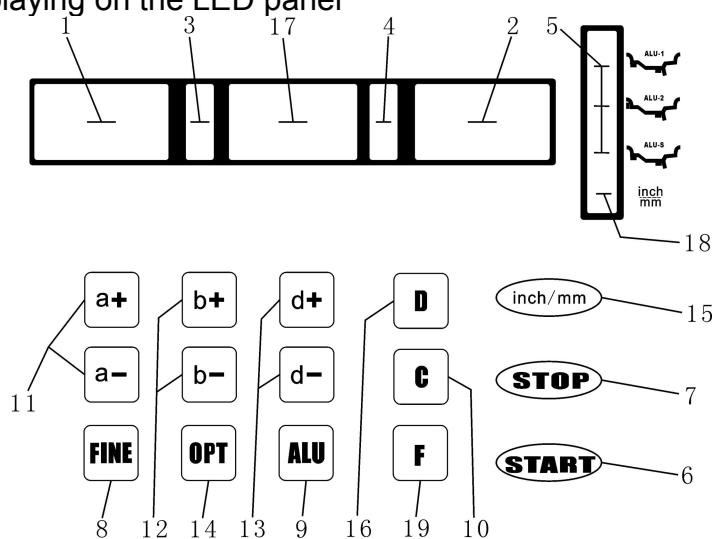


Figure 5-1

Part name of panel and signification

- 1-Digital readout, amount of imbalance, inside or "DISTANCE" dimension
- 2-Digital readout, amount of imbalance, outside or "DIAMETER" dimension
- 3-Digital readout, position of imbalance, inside
- 4-Digital readout, position of imbalance, outside
- 5-Indicator, "ALU" correction mode selected
- 6-Push button, cycle start
- 7-Push button, emergency and selection of special functions
- 8-Push button, unbalance display pitch and threshold
- 9-Push button, selection of "ALU" mode of correction
- 10- Push buttons for recalculation and self-calibration
- 11- Push buttons, manual DISTANCE (a) setting
- 12- Push buttons, manual WIDTH (b) setting
- 13- Push buttons, manual DIAMETER (d) setting
- 14- Push button, optimization of unbalance and split unbalance
- 15- Selection of dimensions, inch/mm
- 16- Push button, self-diagnostics, self-calibration and split unbalance
- 17- Digital readout, "STATIC" unbalance or "WIDTH" dimension
- 18- Indicator of dimensions in mm
- 19- Selection, "STATIC" or "DYNAMIC" correction

N.B. Only use the fingers to press push buttons. Never use the counterweight pincers or other pointed objects.

6. Installation and Demounting of the Wheel

6.1 Checking the wheel

The wheel must be clean, none sand or dust on it, and remove all the primal counterweights of the wheel. Check the tyre pressure whether up to the rated value. Check positioning plane of rim and mounting holes whether deformed.

6.2 Installing the wheel

6.2.1 Select the optimal cone for the center hole when there is center hole on the rim.

6.2.2 Two ways of installing the wheel: A. positive positioning; B. negative positioning.

6.2.2.1 Positive positioning (refer to figure 6-1):

Positive positioning is commonly used. It operates easily, and it is applicable for various rims of common steel structure and thin duralumin structure.

6.2.2.2 Negative positioning (refer to figure 6-2):

Negative positioning is used to ensure the inner hole of steel rim and main axis is positioning accurately when the outside of wheel deforming. Apply for all the steel rims, thick steel rims especially.

6.2.3 Install wheel and cone on main axis. Ensure the cone can clamp the wheel before screwing handle. Wheel can rotate after screwing down.

6.3 Demounting the Wheel

6.3.1 Demount the handle and cone.

6.3.2 Put the wheel up, and then take it down from main axis.

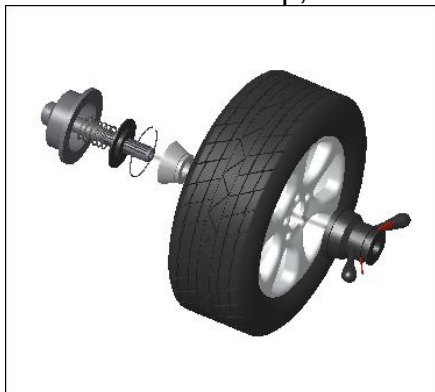


Figure 6-1



Figure 6-2

Note: do not slip wheel on main axis to prevent main axis from scuffing while installation and demounting the Wheel

7. The input methods of date of rim

7.1 The power-on state of the machine

After the power-on of the machine, it starts initialization automatically. The initialization will be finished after two seconds. The machine enter natural dynamic (clamp counterweights on the correction plane of the both edged sides of rim) mode automatically, as in Figure 7-1, ready for input date of rim.

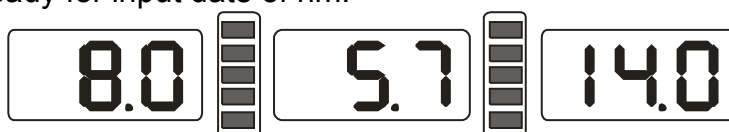


Figure 7-1

7.2 Date of wheel input method for normally dynamic balance mode

7.2.1 After the power-on of the machine, it enter the normal balance mode

7.2.2 Input date of rim:

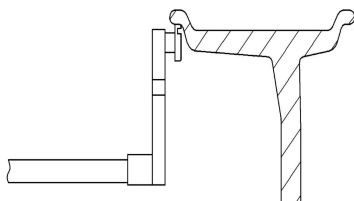


Figure 7-2

Rotate the measure scale, pull the scale head to inside concave on the edge of the rim, etc figure 7-2, first LED all off, then display ect figure 7-3, waiting put back scale.

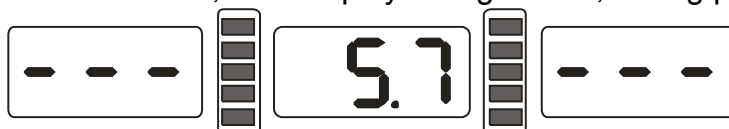


Figure 7-3

When scale back to zero position, LED display value is date of rim.

7.2.3 When measure values with fact value of rim differ, you need self-calibration of scales then measure again or manual input date of rim.

7.2.4 Input date of rim width

Use manual plastic width measure scale, measure the width value of rim then press b+ or b- key input date by manual.

7.3 The date input method of ALU-S mode:

Normally from normal mode change to other mode, do not need input rim date again, just press ALU key change to you select mode in directly is fine, only ALU-S mode have special input method, ALU-S mean is special mode, include below two kind modes (Figure 7-6):



Figure 7-6

ect figure 7-7 or 7-8, Rotate the measure scale, pull the scale head to inside of the rim(FI)(this position is your select to stick balance weight on the inside), measure distance(a) and diameter (dI) of inside of the rim, ect figure 7-9, then continue rotate measure scale, pull the scale head to outside of rim(FE) (this position is your select to stick balance weight on the outside), measure distance(aE) and diameter (dE) of outside of the rim, ect figure 7-10, enter ALU-S mode.

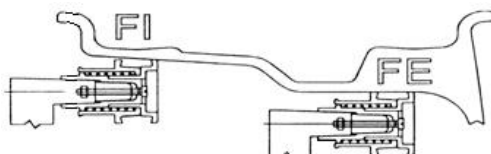


Figure 7-7

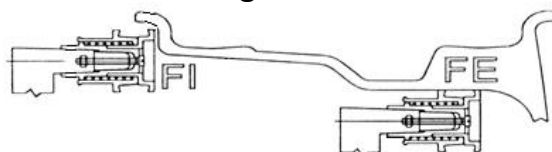


Figure 7-8

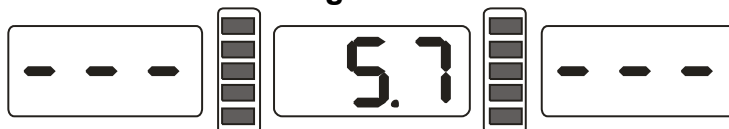


Figure 7-9**Figure 7-10**

When scale back to zero position, LED will be display aE and dI value. Press a+, a- can be adjust a value; press b+, b- can be adjust aE value; press d+, d- can be adjust dI value; press FINE key LED will display dE value, hold press FINE key and press d+, d- adjust dE value.

8. Calibration of measure scale

The measure scale was labeled before ex-factory, but the labeled value may vary because of transportation. Therefore, users can label it themselves before using the balancer.

After the power-on, the initialization is finished. Thus, users can label the measure scale

8.1 Calibration of rim distance scale

8.1.1 Press and hold STOP key and press FINE key, ect 8-1, press STOP key or C key to exit.

**Figure 8-1**

8.1.2 Move scale to 0(zero) position, press ALU key, ect 8-2, press STOP key or C key to exit.

**Figure 8-2**

8.1.3 Move scale to 15, press ALU key, ect 8-3, end of self-calibration, put back measure scale.

**Figure 8-3**

8.2 Calibration of diameter scale

8.2.1 Installation middle size tyre on principal axis, press and hold STOP key, and press OPT key, ect 8-4, press STOP key to exit;

**Figure 8-4**

8.2.2 Press d+ or d- adjust value to current rim diameter value, press ALU key, ect 8-5;



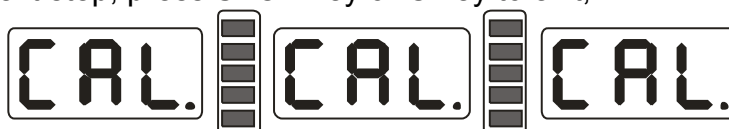
Figure 8-5

8.2.3 Move scale, put scale head on the inside edge of rim ect 7-2, press ALU key, ect 8-3, self-calibration end, put back scale.

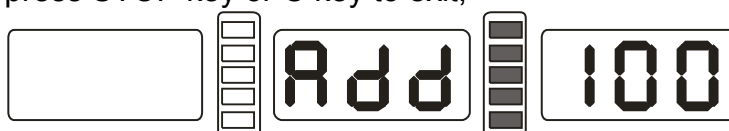
9. The Self-calibrating of Dynamic Balancer

The self-calibrating of dynamic balancer was finished before ex-factory, but the system parameter may vary because of long-distance transportation or long-term use, which may cause error. Therefore, users can make self-calibrating after a period of time.

- 9.1 After the power-on of the machine, the initialization is finished ect 7-1, install a can clip counterweight and comparatively balanced middle size tyre, follow step 7 input date of rim;
- 9.2 Press D key and C key, ect figure 9-1, (lay down protect hood), press START key, going to next step, press STOP key or C key to exit;

**Figure 9-1**

- 9.3 After axis stop, ect figure 9-2, (open protect hood) clip a 100 gram counterweight on anywhere of outside of rim, (lay down protect hood) press START key, going to next step, press STOP key or C key to exit;

**Figure 9-2**

- 9.4 After axis stop, ect 9-3, calibration end. Demount tyre, now balancer ready to work.

**Figure 9-3**

NB: when you doing self-calibration, input date of rim must be correct, 100 gram counterweight must be correct, otherwise self-calibration result will be wrong, wrong self-calibration will be make balancer measure precision decline.

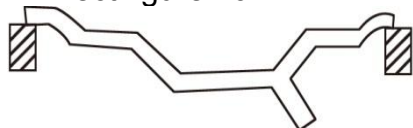
10. The Balancing Operation of Wheel

10.1 Balance mode change operation

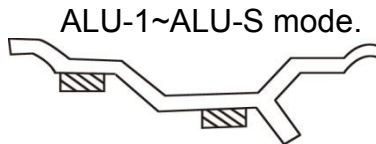
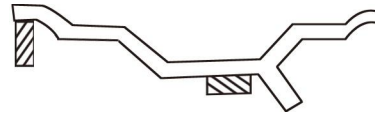
- 10.1.1 Dynamic and static balance mode switch: press F key.

Dynamic balance mode: clip counterweight on inside and outside edge of rim (balancer tolerant normal balance mode), ect figure 10-1.

ST mode: static balance measure mode, stick counterweight on middle of rim, ect figure 10-2.

**Figure 10-1****Figure 10-2**

- 10.1.2 ALU balance mode switch key (ALU key), switch CPU system between

**Figure 10-3****Figure 10-4**

ALU-1 mode: stick counterweight on inside and outside of spoke of rim, ect 10-3.

ALU-2 mode: clip counterweight on inside edge of rim, stick counterweight on outside (inside of spoke), etc figure 10-4.

ALU-S mode: This is a special balance mode (include ALU-1 and ALU-2), for Counterweight split and Hidden-Stick Mode.

Counterweight split and Hidden-Stick Mode:

ALU-S mode, if the outside (inside of spoke) counterweight position between of two spokes, ALU-S mode can split the counterweight for two. The divided counterweight, just place on the back of two spokes near the former counterweight, in order to hide the counterweight. ect figure 10-5.

**Figure 10-5**

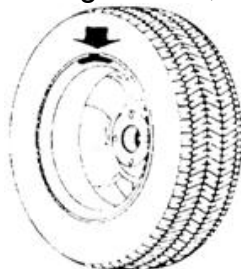
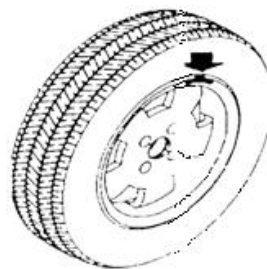
10.2 Normal balance mode operation process

10.2.1 Follow figure 7.2 input date of rim.

10.2.2 lay down protect hood and press START key, wheel spinning, after stop two side LED display wheel both side imbalance weight, when middle LED display OPT, can choice imbalance optimize;

10.2.3 Slowly spin wheel, when inside counterweight position indicator light (figure 5-1(3)) all on, clip correspond counterweight on 12 o'clock position on inside of rim, ect figure 10-6;

10.2.4 Slowly spin wheel, when outside counterweight position indicator light (figure 5-1(4)) all on, clip correspond counterweight on 12 o'clock position on outside of rim, ect figure 10-7;

**Figure 10-6****Figure 10-7**

10.3 Static balance (ST) operation process

10.3.1 Move measure scale to middle of rim, and measure the data of rim;

10.3.2 Press F key, switch to static balance (ST) mode.

10.3.3 lay down protect hood and press START key, wheel spinning, after stop

middle LED display wheel static imbalance weight, when middle LED display OPT, can choice imbalance optimize;

- 10.3.4 Slowly spin wheel, when both side counterweight position indicator light (figure 5-1(3), (4)) all on, stick counterweight on 12 o'clock position middle of rim. ect figure 10-8.



Figure 10-8

10.4 The operation process of M1 to M2 mode

- 10.4.1 Follow 7.2 input date of rim;
- 10.4.2 Press ALU key, switch to correspond balance mode.
- 10.4.3 lay down protect hood and press START key, wheel spinning, after stop two side LED display wheel both side imbalance weight, when middle LED display OPT, can choice imbalance optimize;
- 10.4.4 Slowly spin wheel, when inside counterweight position indicator light (figure 5-1(3)) all on, follow different mode clip or stick counterweight on 12 o'clock position inside of rim.
- 10.4.5 Slowly spin wheel, when outside counterweight position indicator light (figure 5-1(4)) all on, follow different mode clip or stick counterweight on 12 o'clock position outside of rim.

10.5 ALU-S balance mode operation process

★ Process of manual stickup counterweight

- 10.5.1 Follow figure 7.3 input date of rim.
- 10.5.2 lay down protect hood and press START key, wheel spinning, after stop two side LED display wheel both side imbalance weight, when middle LED display OPT, can choice imbalance optimize;
- 10.5.3 Slowly spin wheel, when inside counterweight position indicator light(figure 5-1(3)) all on, if is follow figure 7-8 input date of rim, next step follow 10.2.3 clip counterweight, if Notec figure 10-9 of left side show, stick counterweight on 12 o'clock position on inside of rim.
- 10.5.4 Slowly spin wheel, when inside counterweight position indicator light(figure 5-1(4)) all on, stick correspond counterweight on 12 o'clock position on outside of rim, ect figure 10-9 right side;

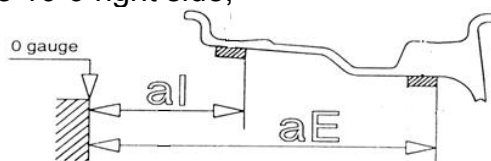


Figure 10-9

★ Process of automatic stickup partition Counterweight

- 10.5.5 Follow figure 7.3 input date of rim
- 10.5.6 lay down protect hood and press START key, wheel spinning, after stop two side LED display wheel both side imbalance weight, when middle LED display OPT, can choice imbalance optimize;
- 10.5.7 If is follow figure 7-8 input date of rim, follow 10.2.3 clip counterweight on inside edge of rim. Press STOP key and ALU key, all middle of LED display - - -, follow 10.5.9 stick counterweight on outside of rim.
- 10.5.8 If is follow 7-7 input date of rim, press STOP key and ALU key, all middle of LED display - - -,insert correspond counterweight into the scale head slot,

slowly spin wheel, when inside counterweight position indicator light (figure 5-1(9)) all on, draught scale, when inside counterweight position indicator light (figure 5-1(3)) all on, slowly pull out measure scale and middle of LED display as figure 10-10, then turn the scale, stick counterweight on the rim, ect figure 10-12;



Figure 10-10

10.5.9 Insert correspond counterweight into the scale head slot, slowly spin wheel, when outside counterweight position indicator light (figure 5-1(4)) all on, draught scale, when inside counterweight position indicator light (figure 10-11) all on, turn scale, stick counterweight on the rim, ect figure 10-12;

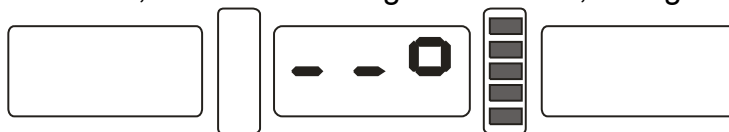


Figure 10-11

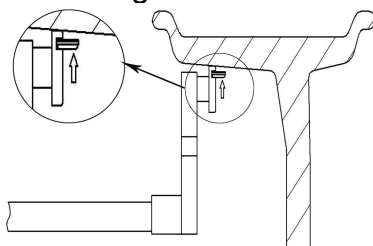


Figure 10-12

NB: when use scale to stick counterweight automatically, make sure LED must display - - -, if not, do not move scale, press STOP and ALU key, then middle LED display - - -, now you can move scale for automatically sticking counterweight.

10.6 The of counterweight hide-stick mode

Counterweight Hidden-Stick mode is only available in ALU-S mode, this mode can split this position into two parts and make these two new positions hide behind the spoke, so you can achieve this hide purpose. This balance mode only available in ALU-S mode.

In 10.3 operation process, if outside counterweight not hide behind the spoke, but you still want it hide behind the spoke, you can follow below operation:

10.6.1 Press a+ key, back to figure 7-1 interface, press D key and OPT key, display input numbers of spoke (figure 10-13), press b+ or b- key input spoke numbers, press D key and OPT key to save and return, slowly turn the wheel, turn the near by imbalance position of spoke on the top (12 o'clock), press D and OPT key to enter hide stick mode, and then press D and OPT key to exit;

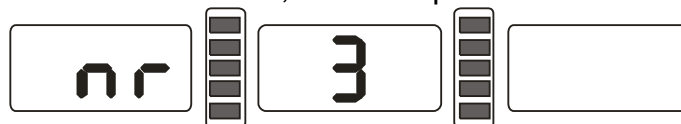


Figure 10-13

★ Split manual stick operation

10.6.2 Stick inside operation same 10.5.3;

10.6.3 Slowly spin wheel, when outside counterweight position indicator light (figure 5-1(4)) all on, stick counterweight on 12 o'clock position outside of rim, ect

figure 10-9 right side;

- 10.6.4 Slowly spin wheel again, when outside counterweight position indicator light (figure 5-1(4)) all on, find second position, stick counterweight on 12 o'clock position outside of rim, ect figure 10-9 right side;

★ Split automatic stick operation

- 10.6.5 When stick inside, operation same 10.3.7 and 10.3.8

- 10.6.6 Slowly spin wheel, when outside counterweight position indicator light (figure 5-1(4)) all on, find first position, draught scale, when outside counterweight position indicator light (figure 10-11) all on, turn scale, stick counterweight on rim (ect figure 10-12);

- 10.6.7 Slowly spin wheel, when outside counterweight position indicator light (figure 5-1(4)) all on, find second position, draught scale, when outside counterweight position indicator light (figure 10-11) all on, turn scale, stick counterweight on rim (ect figure 10-12).

NB: when use scale to stick counterweight automatically, make sure LED must display - - -, if not, do not move scale, press STOP and ALU key, then middle LED display - - -, now you can move scale for automatically sticking counterweight.

10.7 Recalculation

Before wheel balance testing, sometimes may forget input current date of rim, you can after wheel balance testing input date of rim reintroduce, at this time do not need press START key, just press recalculation key (C) system will follow new date of rim calculation imbalance measure. Even interface currently display imbalance value, press C key, can be check currently input date of rim.

11. Imbalance optimize

If wheel imbalance value over 30 gram, system will be display "OPT", clue on process imbalance optimize.

Imbalance optimize have two operation method:

11.1 Already display balance value

If already finish balance testing, when you need process imbalance optimize, press OPT key, display ect figure 11-1;



Figure 11-1

Use chalk mark a reference point on the flange plane and rim and tyre, use tyre changer exchange rim and tyre 180°; Reinstall wheel on the balancer and make sure mark of reference point between the flange plane and rim must be on the same position. press START key, display ect figure 11-2;

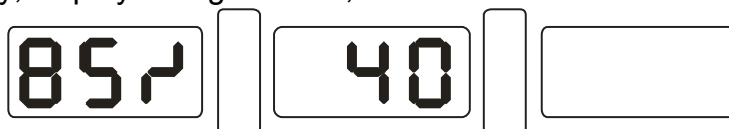


Figure 11-2

ect Figure 11-2, left LED display percent of optimize, if before optimize static value is 40 gram, optimized is 85%, so after optimize static value only 6 gram remain ($15\% \times 40\text{gram} = 6\text{gram}$);

Slowly spin wheel by hand, when both side two of position indicator light

flashing (figure 11-3), use mark pen make a mark on the tyre;



Figure 11-3

Slowly spin wheel by hand again, when both side one of middle position indicator light flashing (figure 11-4), use mark pen make a mark on the rim;



Figure 11-4

Remove the wheel from balancer, use tyre changer remove tyre from rim, make tyre and rim marks together, use tyre changer fit tyre on the rim, optimize complete.

11.2 Before testing of power on, also process imbalance optimize directly

Turn on the power, installation wheel, press OPT key, left side display OPT, press START key, display ect figure 11-1, then follow 11.1 step to operation. Press STOP key to stop operation.

12. Gram-Oz conversion operation

This operation for counterweight maund conversion (gram-Oz).

12.1 Press STOP key and a+ key or a-, display ect figure 12-1, denotation currently maund is gram.

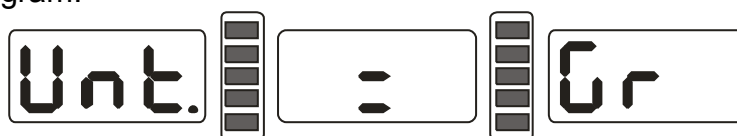


Figure 12-1

12.2 Press b+ or b- key, display ect figure 12-2, denotation currently maund is Oz;



Figure 12-2

12.3 Press b+ or b- key for switch maund between gram and Oz;

12.4 Press a+ key to save setup and exit.

13. INCH and MM conversion operation

This operation for input dimension B value and D value conversion (INCH-MM).

Press INCH/MM key, long measure of LED 6 display value B and LED 7 display value D can be change between INCH/MM. if in the LED display window right underside have a radix point mean currently long measure is INCH, if not, long measure is MM.

14. Protect hood function and protect setting

This function can be setting closing protect hood motor start or lay down protect hood properly, then press START key to start motor.

Press STOP key and C key, ect figure 14-1, right display currently state, when display ON, denotation function on, when display OFF, denotation function off.

Press b+ or b- key for switch protect hood function between "ON" and "OFF";

Press a+ key to save currently settings and exit.

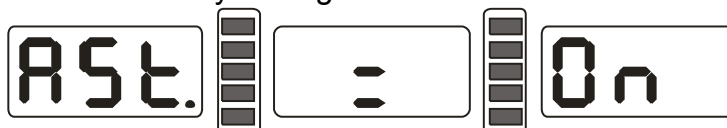


Figure 14-1

15. Other function settings

15.1 Minimum value display settings

After select minimum value display ,even wheel imbalance value less than setting value, result will be display 0(zero),press FINE key, will be display fact imbalance value. Press STOP and D key, ect figure 15-1,denotation imbalance value less than 5 gram, display is 0g(zero gram),press b+ or b- key setting minimum value, are 5 and 10 and 15gram three distinctions. Press a+ key to save settings and enter next step;



Figure 15-1

15.2 Key-tone clue on function settings

This function can turn on or turn off key-tone, even turn on this function, every time press key, system will be emit a “di” of tone, even turn off this function, press key not tone. From 15.1 press a+ enter settings, ect figure 15-2,right side display ON denotation function on, display OFF denotation function off. Press b+ or b- key, function will be switch between “ON” and “OFF”. Press a+ key to save settings and enter next step;



Figure 15-2

15.3 Display monitor brightness settings

This function will follow use environment and user need, setting display brightness. From 15.2 press a+ enter settings, ect figure 15-3,right side display brightness grade, altogether is 8 levels, level 1 is darkest, level 8 is brightest, default is level 4.press b+ or b- key, function will be switch brightness levels. Press a+ key to save settings and enter next step;



Figure 15-3

16. Machine self test function

This function will be check all kinds input signal whether right working and provide gist for trouble analyses.

16.1 LED and indicator light check

Press D key, whole LED and indicator light all turn on, this function can be check LED or indicator light whether appear damage, press C key to exit, after about 5 seconds display ect figure 16-1,enter position sensor check. Press C key to exit.



Figure 16-1**16.2 Position sensor signal check**

This function can be check position sensor and principal axis and main board circuit whether damage.

Slowly turn principal axis, right side LED display number appear change, clockwise turn, and value increase. Anticlockwise turn, value minish, correct value change is between 0 to 63.press ALU key, enter distance sensor check. Press C key to exit.

16.3 Distance sensor signal check

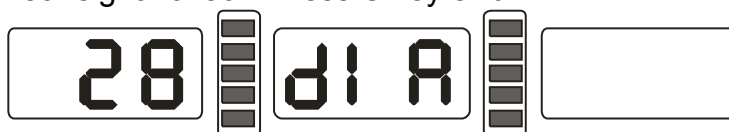
This function can be check distance sensor and main board signal circuit whether damage.

From 16.2 press ALU key enter, ect figure 16-2, move measure scale, value will be change, value follow move more increase. Press ALU key, enter diameter sensor signal check. Press C key exit.

**Figure 16-2****16.4 Diameter sensor signal check**

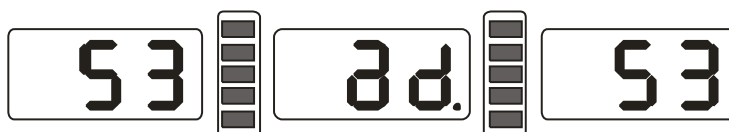
This function can be check diameter sensor and main board signal circuit whether damage.

From 16.3 press ALU key enter, ect figure 16-3, turn measure scale, value will be change, anticlockwise turn, value increase, clockwise press turn, value minish. Press ALU key, enter press sensor signal check. Press C key exit.

**Figure 16-3****16.5 Press sensor signal check**

This function can be check press sensor and main board signal circuit and power supply whether damage.

From 16.4 press ALU key enter, ect figure 16-4, gently press principal axis, right and left LED display value will be change, press ALU key to enter press sensor signal check. Press C key exit.

**Figure 16-4****17. Safety Protections and Trouble Shooting****17.1 Safety protection**

17.1.1 Under the circumstance of operation, if the machine does not operate normally, press STOP key, the rotating wheel will stop immediately.

17.1.2 If hood is not laid down, press START key, the wheel will not rotate.

17.1.3 Under the circumstance of operation, if hood open, the rotating wheel will stop immediately.

17.2 Trouble shooting

17.2.1 After press START key, principal-axis not spinning, LED display Err-1.check

- motor, computer board and cable connections;
- 17.2.2 After press START key, principal axis rotate, LED display Err-1. check position sensor, computer board and cable connections;
- 17.2.3 If balance test already end, balancer still rotates for a long time, does not brake, check brake resistance, power supply board, computer board and cable connections;
- 17.2.4 If turn on the power LED display ect figure 17-1, need measure scale self-calibration or adjust a-value sensor or replacement new sensor;

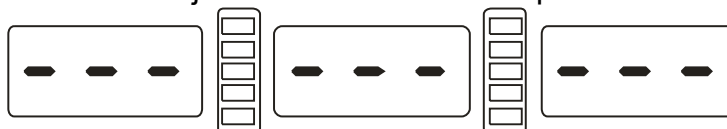


Figure 17-1

- 17.2.5 Automatic measure rim, advert display date whether difference with trim size of rim, if difference, need measure scale self-calibration;
- 17.2.6 If turn on the power have not display, check power switch indicator light whether working, if not, check power supply first, then check power supply board and computer board and cable connections;
- 17.2.7 Precision not nicety, commonly hurtful not from balancer, may wheel installation wrong or counterweight heft not nicety or 100 gram counterweight for balance self-calibration not nicety, must be always keep manufactory's 100 gram counterweight safety for self-calibration only.
- 17.2.8 Date not astaticism, repeat difference, commonly hurtful not from balancer, may wheel installation wrong or floor not messiness not level off, may as well fixation balancer by bolt. Sometimes not connection earth wire may appear this phenomenon.
- 17.2.9 If add weights many times, the tire still can not be balanced, it is possible the operator did find the correct unbalanced position and weights were not added on the correct position. Follow the instructions to do the self-calibration once. If still can not solve the problem, check it as following ways: 1)put down the protective cover, start the machine to test the tire; 2)turn the tire slowly by hand, to find the lateral unbalanced position; 3)add a 100g weight on the lateral position of rim(12 o' clock position); start the machine to test the tire, turn the tire by hand slowly, to find the lateral unbalanced position; check if the position of 100g weight is on the position of 6 o' clock(the bottom position); If no, it means the parameters of the machine has changed, please contact the dealer or manufacturer to solve it.

Hint: check precision right method:

Input right date of wheel(a. b. d value),consult instruction do a self-calibration, press START process balance operation, note down date of first time, clip 100 gram counterweight on the outside edge of wheel(when outside indicator light all on is top zenith position),press START key again process balance operation, this date of outside display addition date of first time, should amount 100 ± 2 ,manually slowly turn the wheel, when light of outside all on, check 100 gram counterweight whether at 6 o'clock position, if not amount 100 gram or 100 gram counterweight not at 6 o'clock position, indicate balancer precision have problem, if amount is 100 gram, follow same method check inside, check inside whether amount is 100 gram and at 6 o'clock.

18. Laser

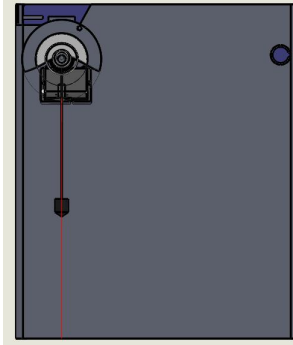
18.1 Laser Settings

Before enter the laser settings, please ensure that the machine is working properly, or the laser program can not be set arbitrarily;

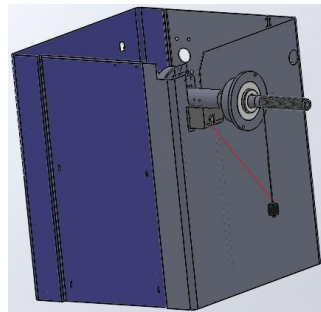
First, install a tire with iron rim, and input a, b, d value correctly, then remove the tire;

Press "M" key to enter the menu interface. Press "M" key again and select "Laser Setting", and press "E" key to enter the laser setting program;

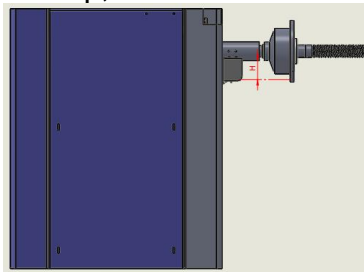
18.1.1 Hang an object on the inside of the shaft, and adjust the position of laser box, to make the laser point towards the bottom (factory adjusted, you can skip), then press "ALU" to the next step;



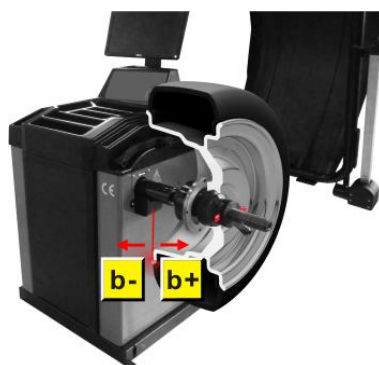
18.1.2 Hang an object on the outside of the shaft, and adjust the screw at the bottom of the laser box, to make the laser point and the outer vertical line overlapped (factory adjusted, you can skip), then press "ALU" to the next step;



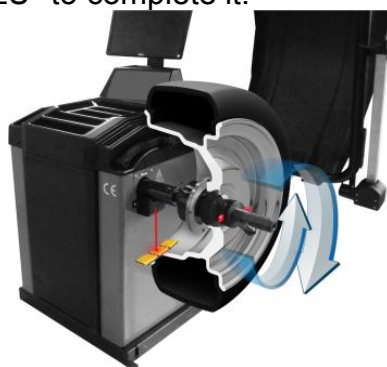
18.1.3 Measure the distance between the laser and the center of the shaft, and press "b +" or "b-" to input the H value (factory adjusted, you can skip), then press "ALU" to the next step;



18.1.4 Press "b +" or "b-" to align the laser point to the inner edge of the rim (factory adjusted, you can skip), then press "ALU" to the next step;

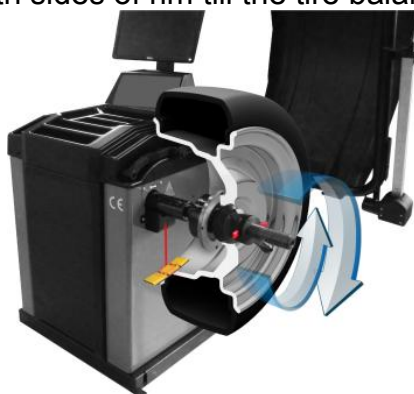


- 18.1.5 Install the tire with iron rim, put down the protective cover (press "START" button), and wait for the end of the balance test;
- 18.1.6 Turn the tire by hand to find the imbalance point, add 100g weight on the above of the tire, put down the protective cover (press "START" button), and wait for the end of the balance test;
- 18.1.7 Turn the tire by hand, to make the center of the weights align to the laser point, and press "ALU" to complete it.



18.2 The Usage of Laser

The laser can only be used in the ALU-S mode. After the balance finish, press "T" key and rotate the tire by hand. When find the unbalanced position, the laser will point to the position where the weights can be pasted. Make the left side (the middle position) of the weight align to the laser point, then paste the weight on the rim. Do the same operation on both sides of rim till the tire balance finish.



19. Maintenance

19.1 The daily maintenance of non- professionals

Before the maintenance, please switch off the power-supply.

19.1.1 Adjust the tension of the belt.

19.1.1.1 Dismantle the hood.

19.1.1.2 Unlash the screw of the motor, move the motor till the belt's tension is proper, and emphatically press the belt downwards about 4mm.

19.1.1.3 Tighten the screw of the motor and cover the hood.

19.1.2 Check whether the wire of electricity part connects reliably.

19.1.3 Check whether the pressed screw of the main axis is loose.

18.1.3.1 Locking nut can not fix wheel tighten on main-axis

18.1.3.2 Use hexagonal wrench to tighten the pressed screw of the main-axis.

19.2 The maintenance of professionals

The maintenance of professionals can only be carried out by the professionals from the factory.

19.2.1 If the imbalance value of tested wheel has obvious errors and does not improve after self-calibrating, this proves the parameter in the machine has altered, so the user should ask for professionals.

19.2.2 The replacing and adjustment of pressure sensor should be operated according to the following methods, and the operation should be carried out by professionals.

The steps are as follows:

1. Unlash the No.1, 2,3,4,5 nuts.

2. Dismantle the sensor and nut.

3. Replace No.6, 7 the sensor organ.

4. Install the sensor and the nut according to the Figure 18-1. (Pay attention to the sensor's direction.)

5. Tighten No.1 nut emphatically.

6. Tighten the No.2 nut to make the main axis and the flank of cabinet, and then emphatically tighten the No.3 nut.

7. Tighten the No.4 nut (not too emphatically), then tighten No.5 nut.

19.2.3 The replacing of circuit board and the organ on it should be carried out by professionals.

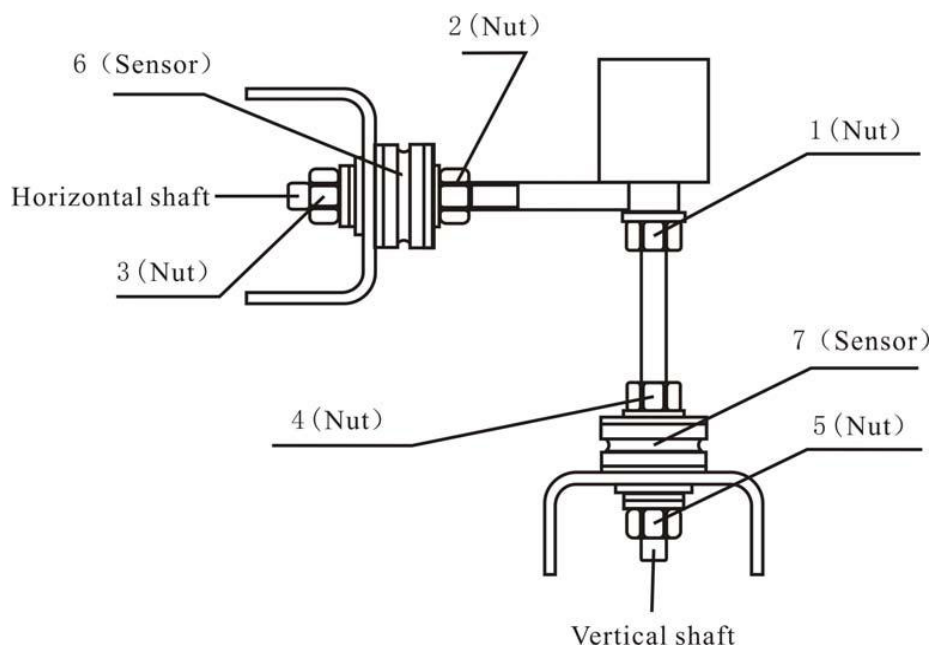


Figure 19-1

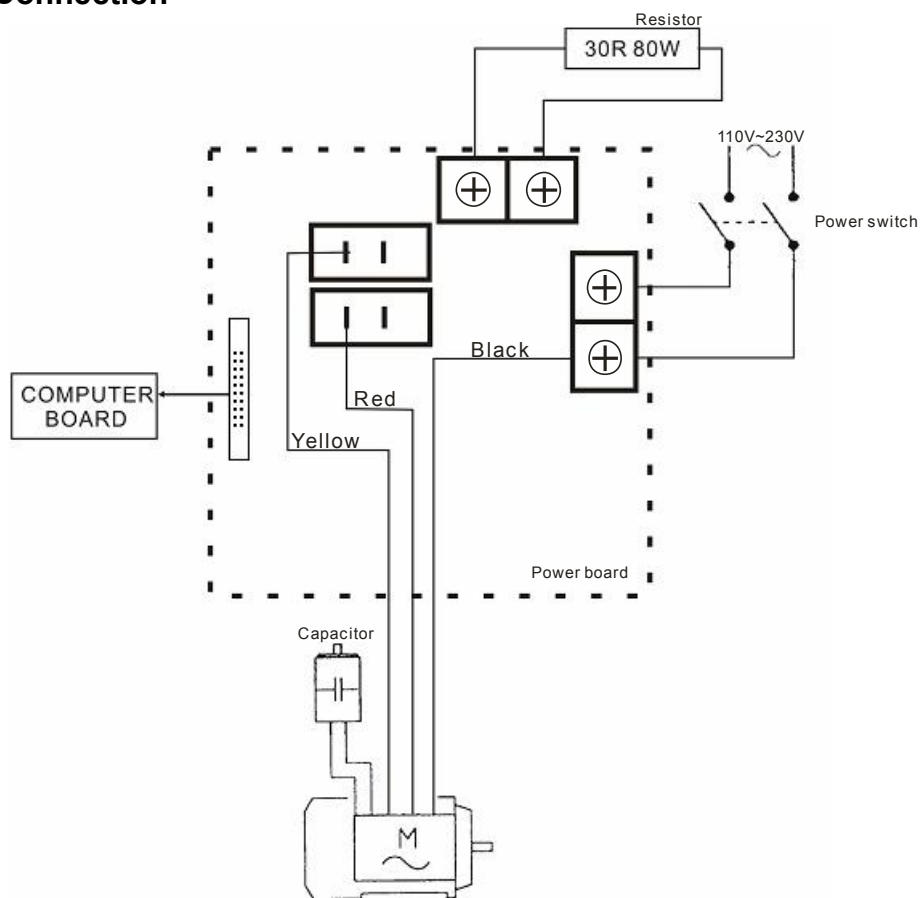
20. Trouble-error code list

When balancer display hint of error, can follow consult below list to remove the trouble:

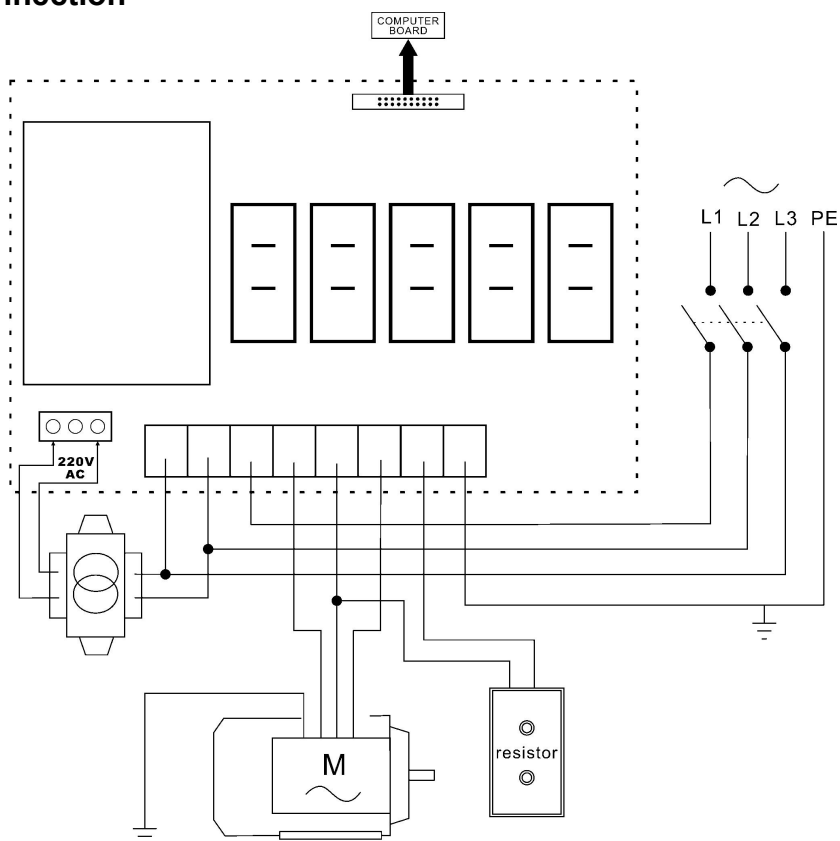
Code	meanings	cause	remedy
Err 1	principal axis not spin or have not spin signal	1.motor fault 2.position sensor fault 3.power supply board fault 4.computer board fault 5.connection-peg untouched	1.change motor 2.change position sensor 3.change power supply board 4.change computer board 5.check cable connections
Err 2	The rotation is lower than 60r/min	1. position sensor fault 2. wheel not impacting or weight too light 3. motor fault 4. driving belt too loose or too tighten 5. computer board fault	1. change position sensor 2.repeat impacting wheel 3. change motor 4.adjust driving belt elasticity 5. change computer board
Err 3	Miscalculation	too high imbalance	Repeat the self-calibration change computer board
Err 4	principal axis Wrong rotation direction	1. position sensor fault 2. computer board fault	1. change position sensor 2. change computer board
Err 5	Protection hood not lay down	1.when press START key, Protection hood not lay down 2. jiggle switch fault 3. computer board fault	1.follow right method operation 2.change jiggle switch 3. change computer board
Err 6	Sensor signal transact circuit not working	1. power supply board fault 2. computer board fault	1. change power supply board 2. change computer board
Err 7	Lose date of interior	1. Incorrect self-calibration 2. computer board fault	1. Repeat the self-calibration 2. change computer board
Err 8	Self-calibration memory fault	1. not put 100 gram on the rim when self-calibration 2. power supply board fault 3. computer board fault 4. press sensor fault 5. connection-peg untouched	1.follow right method repeat self-calibration 2. change power supply board 3. change computer board 4.change press sensor 5.check cable connection

21. Power supply layout diagram

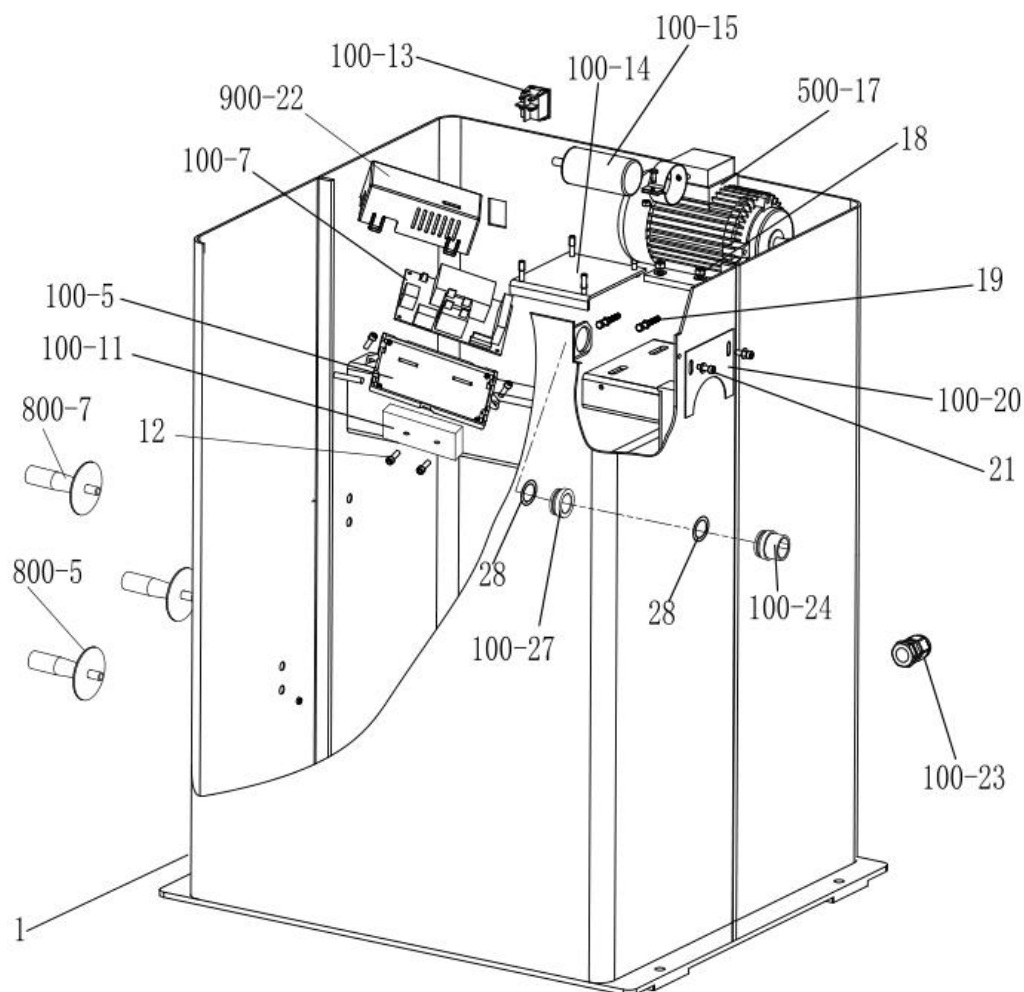
21.1 220 V Connection



21.2 380 V Connection

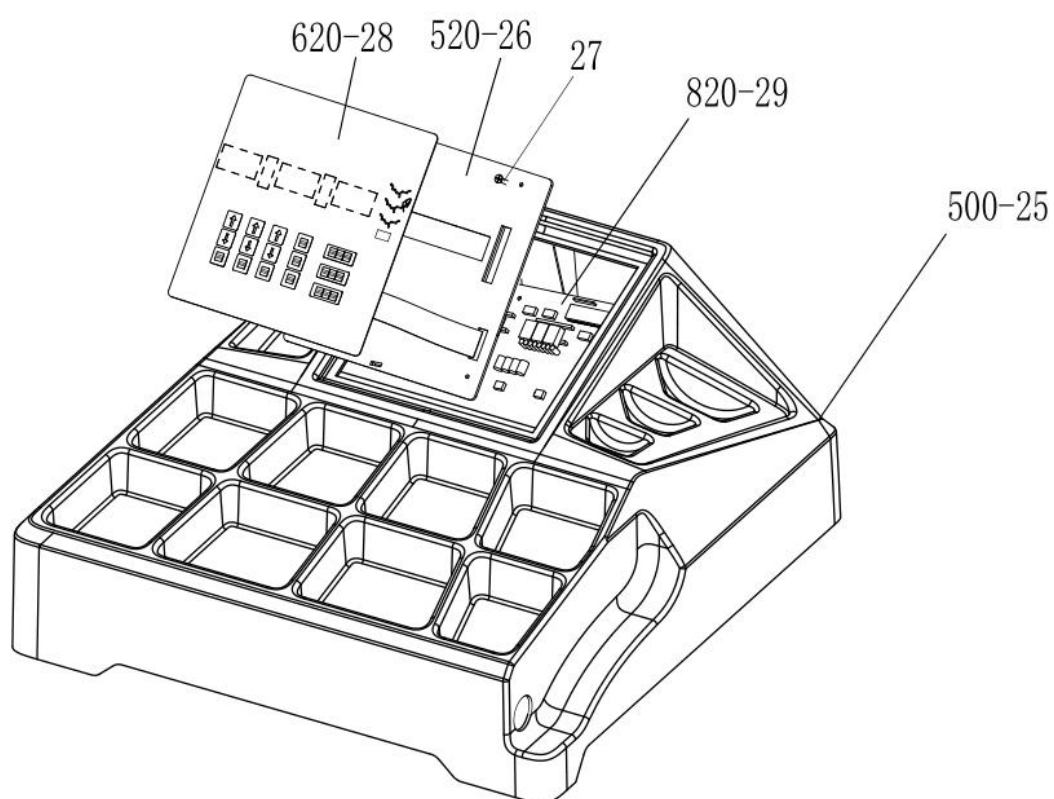


22. Exploded drawings and Spare parts list

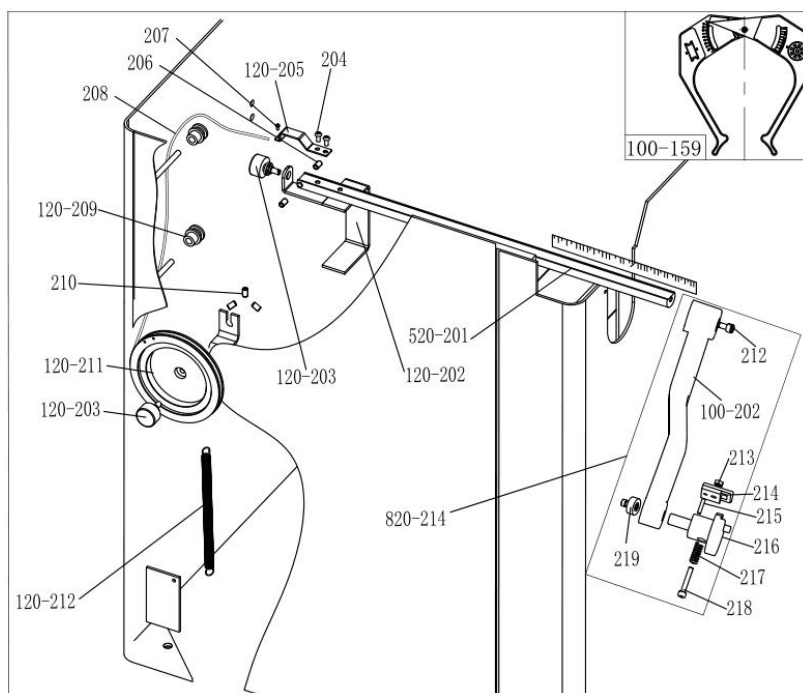


No.	Description	U8 Code	NC Code
1	Body	PX-520-010000-0	JZ08002021399
800-5	Tools hang washer P5-03	B-040-050000-1	JZ07002020948
800-7	Tools hang P5-03	B-024-050251-0	JZ07002020947
100-5	Electric Support Board	P-100-120000-0	JZ07002020944
100-7	Power board	PZ-000-020828-0	JZ02008012184
900-22	Box	P-100-120100-0	JZ07002020945
100-11	Resistor	D-010-100100-1	DD04010002061
12	Screw	B-024-050251-0	FJ04006004341
100-13	Power switch	S-060-000210-0	DD03009001576
100-14	Motor adjust board	PX-100-010920-0	JZ03002021444
100-15	Capacitor	S-063-001500-0	DD04009001880
500-17	Motor	S-051-230020-0	DD01001001178

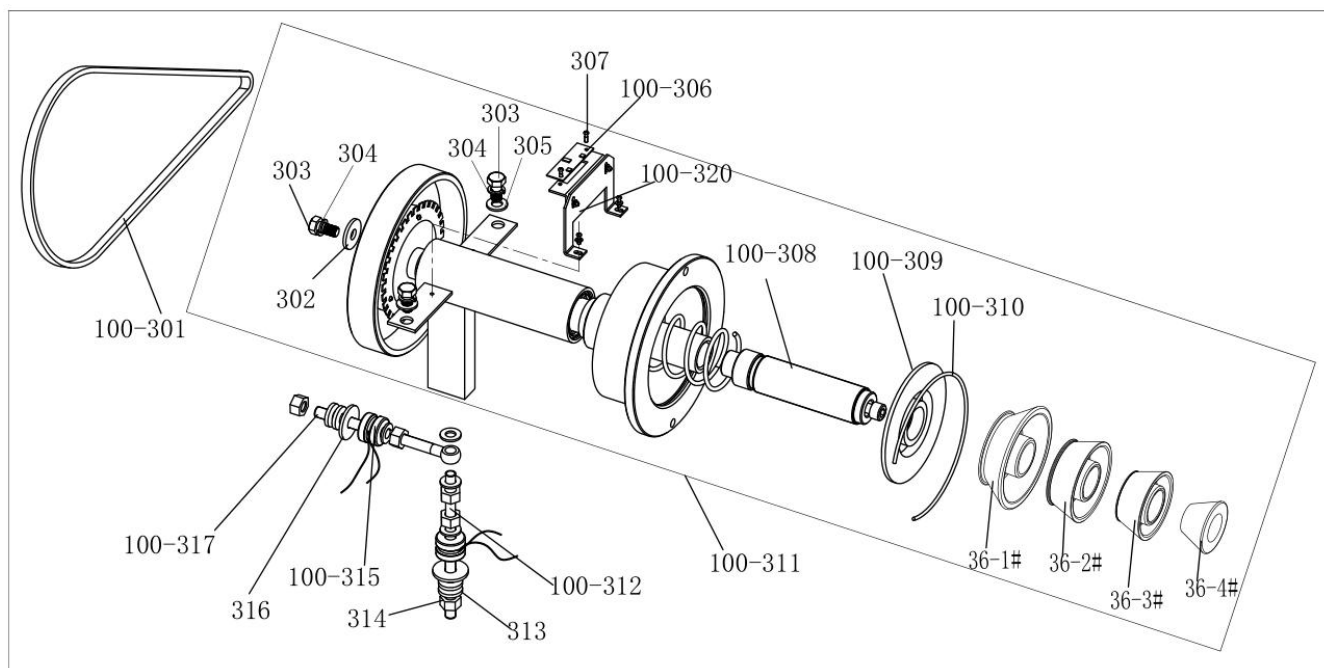
18	Nut	B-004-050001-1	FJ04008004512
19	Screw	B-014-050351-1	FJ04009004849
100-20	Plate	PX-100-110000-0	JZ08002021424
21	Screw	B-024-050061-1	FJ04006004408
100-23	Cable circlip	DS-S-025-000135-0	DD02001001295
100-24	Measuring scale fixing sleeve	P-100-170000-0	JZ07002020964
100-27	Measuring scale fixing sleeve	P-100-170000-A	JZ07002020963
28	Measuring Scale fixing snap spring	P-100-520000-0	JZ09002023428



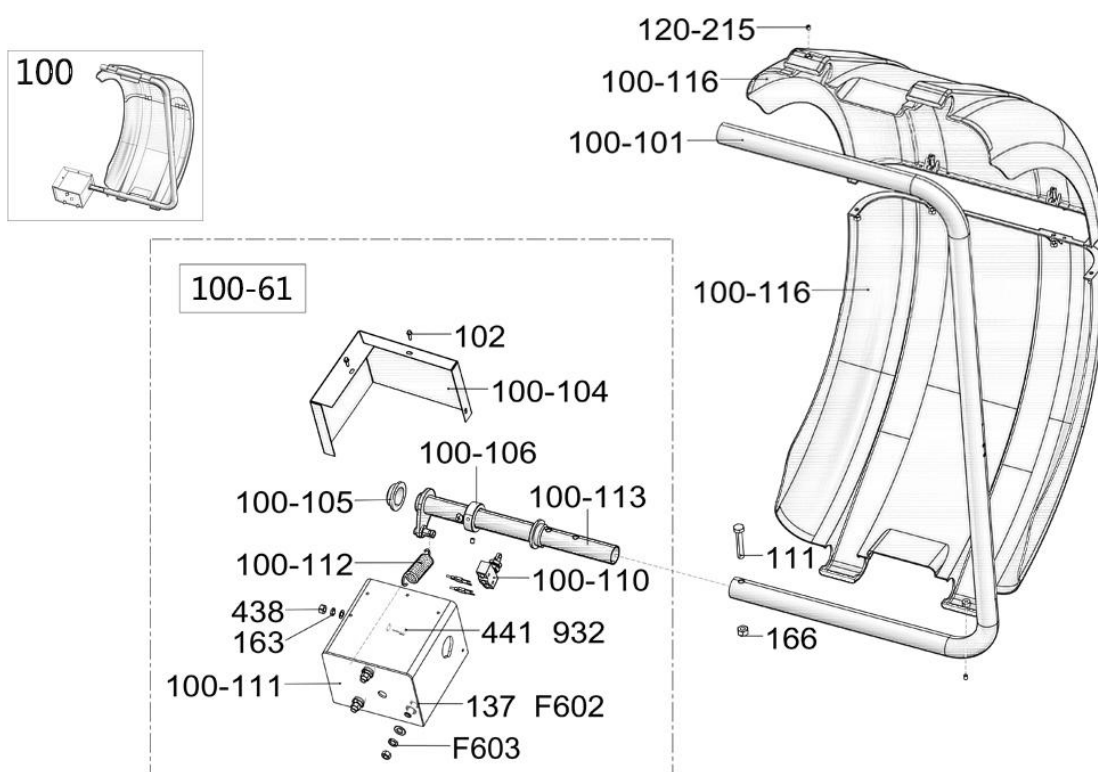
No.	Description	U8 Code	NC Code
620-28	Key board		JZ10001050569
520-26	Display fixed plate	P-520-100000-0	JZ07002020934
27	Screw	B-017-030161-0	FJ04006004306
820-29	Computer board	PZ-000-010820-0	JZ02008012155
500-25	Head with tools-tray	P-500-190000-4	JZ09002023366



No.	Description	U8 Code	NC Code
520-201	Measuring Scale	P-520-090000-0	JZ09002023425
120-202	Heavy	PX-120-240000-0	JZ03002016087
120-203	Gauge sensor	S-132-000010-0	DD04010001969
204	Screw	B-024-050101-1	FJ04006004254
120-205	caliper hook	PX-120-230000-0	JZ03002021386
206	Screw	B-007-050081-0	FJ04006004061
207	Screw	B-024-070301-0	FJ04006004364
208	Steel rope	P-000-030010-0	FJ09001005463
120-209	Pulley	P-120-260000-0	JZ07002020942
210	Screw	B-007-060081-0	FJ04006004063
120-211	Bobbin winder pulley	P-120-250000-0	JZ07002020971
120-212	Spring	P-120-210000-0	JZ09002023435
820-214	Handle Bar	PA-009-082800-0	JZ04002018635
213	Screw	GZBLS-B-011-060161-0,B-010-060161-0	FJ04009004676
100-202	Handle bar 728	P-728-160100-0	JZ07002020965
214	Nut	GZBLS-B-004-040000-0,B-004-040001-1	FJ04008004510
120-215	clamping block	P-828-160902-0	JZ07002020954
216	Pin	B-061-003014-0	FJ06002004998
120-217	base of the stick handle	P-828-160901-0	JZ07002020987
120-218	Button spring	C-221-690400-A	JZ09001022793
219	Screw	B-010-040251-0	FJ04009004638
220	Screw	B-024-050101-1	FJ04006004254
100-59	Caliper	P-000-001008-0	JZ07002020959



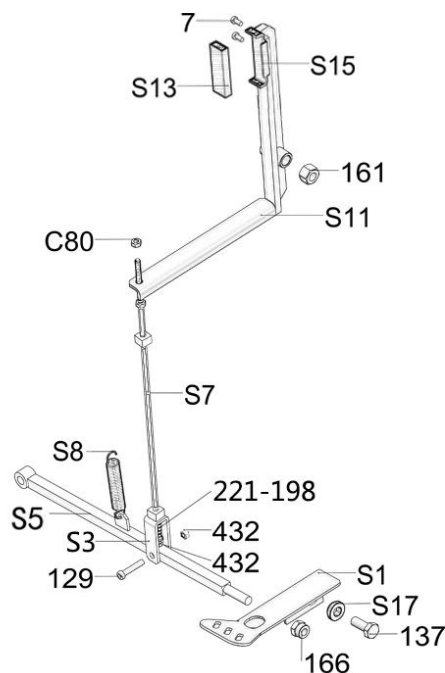
No.	Description	U8 Code	NC Code
100-311	Complete shaft(longer/32/U-8/Tr36)	S-100-000800-0	JZ06002020498
100-308	Thread Tr36	P-100-400000-3	JZ03002015928
100-309	Plastic Lid Tr36	P-100-420000-0	JZ07002020939
100-310	Spring	P-100-340000-0	JZ09002023333
307	Screw	B-024-030061-0	FJ04006004367
100-306	Position Pick-up Board	PZ-000-040100-0	JZ02008012201
302	Washer	B-040-103030-1,B-040-103030-1	FJ04002003804
303	Screw	B-014-100251-1	FJ04009004742
304	Washer	B-050-100000-0	FJ04002003773
305	Washer	B-040-112020-1	FJ04002003800
100-323	Support of Position Pick-up Board 100	P-800-220000-1	JZ09002023341
36-1#	1 CONE Tr36,big	S-100-036000-1	JZ03001015878
36-2#	2 CONE Tr36,medium	S-100-036000-2	JZ03001015879
36-3#	3 CONE Tr36,med-small	S-100-036000-3	JZ03001015880
36-4#	4 CONE Tr36,small	S-100-036000-4	JZ03001015881
100-301	Balancer Belt	S-042-000380-0	FJ02003003528
313	Washer	B-048-102330-1	FJ04002003766
314	Nut	B-004-100001-2	FJ04008004487
316	Washer	B-040-104030-1	FJ04002003806
100-312	Double-end sensing screw rod	P-100-080000-0	JZ09002023331
100-315	Sensor Assembly	S-131-000010-0	DD04007001858
100-317	Single-end sensing screw rod	P-100-070000-0	JZ09002023329



No.	Description	U8 Code	NC Code
100-116	Protection Hood Assembly	P-100-200100-0	JZ07002020936
100-104	Cover for protection hood box	PX-100-030000-0	JZ08002021368
F603	Washer	B-050-100000-0	FJ04002003773
163	Washer	B-050-080000-0	FJ04002003792
F602	Washer	B-040-112020-1,B-040-102020-1	FJ04002003800
932	Washer	B-040-081820-1	FJ04002003906
441	Screw	B-010-080201-0	FJ04009004706
131	Screw	GZBLS-B-011-060161-0,B-010-06016	FJ04009004676
	Hex wrench	S-105-000050-0,G-035-050050-0	HG02001028227
100	complete set of protection hood 100	PW-108-010000-0	JZ06002020459
100-110	Micro switch	S-060-000410-0	DD03009001605
	Micro switch wire	PZ-000-180520-1	JZ02008026590
166	Nut	GZBLS-B-001-100000-0,B-001-10000	FJ04008004444
438	Nut	B-004-080001-0	FJ04008004515
432	Nut	GZBLS-B-001-060000-0,B-001-06000	FJ04008004458
111	Screw	B-014-100451-0	FJ04009004777

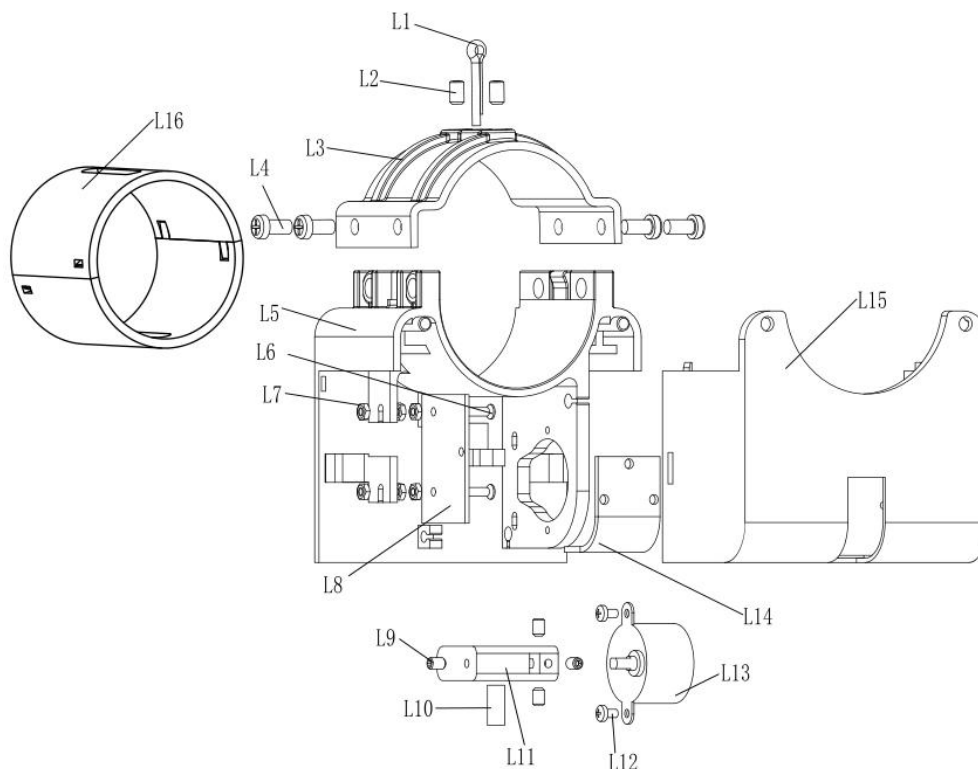
137	Screw	B-014-100251-0	FJ04009004770
102	Screw M4.8*16	B-013-050161-1	FJ04006004050
100-61	Protect axle box assembly 100	PW-108-010000-2	JZ06002020461
100-111	Protection Hood Box	PX-100-020000-0	JZ08002021366
100-101	Protection Hood Rotating Shaft	PX-100-200200-0	JZ09002023655
100-113	Protection hood shaft	PX-800-040000-0	JZ08002021365
	Protection Hood shaft block	P-100-200300-0	JZ09002023297
100-106	Protection hood shaft sleeve	PX-800-050000-0	JZ07002020994
100-112	Protection hood spring	P-800-330000-0	JZ09002023294
120-215	Screw	B-007-060081-0	FJ04006004063
4231	Screw	B-019-350161-0	FJ04006004231
100-105	Sheath	P-800-180000-0	JZ07002020937
	Wrench	PX-100-200400-0	JZ03002016059

S Version Exploded drawings and spare parts list (Pedal brake system)



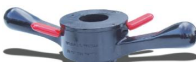
No.	Description	U8 Code	NC Code
S11	Brake lever U100	PX-100-020200-0	JZ03002015963
S15	brake pad mounting seat	PX-100-020700-0	JZ03002015917
S13	brake pad		JZ02008044808
S1	foot pedal	PX-100-020500-0	JZ03002021371
S7	Connecting rod U100	PX-100-020400-0	JZ03002015977
S3	hook	PX-100-020404-0	JZ03002015980
S5	Foot rod Assembly	PX-100-020200-0	JZ03002015963
221-198	Hex locking board spring Φ1.5*9*54	C-221-400000-0	JZ09001022969
129	hex socket head cap screw M6*35	GZBLS-B-011-060351-0	FJ04009004686
166	Nut M10	B-001-10000	FJ04008004444
161	Nut M12	B-001-12000	FJ04008004445
C80	Nut M6	GZBLS-B-004-060000-0,B-004-06000	FJ04008004513
166	Nut M12*1.75*T6	B-004-120071-1	FJ04008004495
432	Nut M6	B-001-06000	FJ04008004458
	Outer hex bolt M12*60	B-014-120601-1	FJ04009038046
137	Outer hex bolt M10*25	B-014-100251-0	FJ04009004770
S17	rotary sleeve	PX-100-020600-0	JZ03002016039
7	screw M5*10	B-024-050101-1	FJ04006004254
S8	Spring U100	P-100-020800-0	JZ09002023304

Laser Version Exploded drawings and spare parts list



No.	Description	U8 Code	NC Code
L1	cotter pin ϕ 4*12	B-070-040012-0	FJ06002005005
L2	Screw M5*8	B-007-050081-0	FJ04006004061
L3	cover	P-579-310900-0	JZ07002020990
L4	Screw M5*10	P-579-310800-0	JZ07002020991
L5	Screw	B-024-050101-1	FJ04006004254
L6	Screw M3*16	B-017-030161-0	FJ04006004306
L7	Nut M3	GZBLS-B-004-030000-0,B-004-030001-1	FJ04008004509
L8	Controller	PZ-000-050579-0	JZ02008012215
L9	Screw	B-007-040061-0	FJ04006004060
L10	Laser	S-054-000002-0	DD04014002121
L11	Laser fixed block	P-579-310200-0	JZ07002020992
L12	Screw	B-024-070301-0	FJ04006004364
L13	Motor	S-053-000001-0	DD01001001162
L14	Glass	P-800-310300-0	JZ09002023515
L15	Box	P-579-310100-0	JZ07002020989
L16	spacer		JZ07002043716

Accessories list

CODE	ITEM	QTY	PHOTO
JZ03001015878	1# CONE (φ36)	1	
JZ03001015879	2# CONE (φ36)	1	
JZ03001015880	3# CONE (φ36)	1	
JZ03001015881	4# CONE (φ36)	1	
JZ06002020443	COMPLETE QUICK RELEASE NUT	1	
JZ03002015928	THREADED SHAFT (φ36)	1	
	MANUAL	1	
JZ03002016059	WRENCH	1	
HG02001028224	HEX WRENCH	1	
HG02001028227	HEX WRENCH	1	
JZ09002023492	STANDARD WEIGHTS 100G	1	
JZ07002020959	CALIPER	1	
JZ09002023563	PLIER	1	
JZ07002020974	PLASTIC LID	1	
JZ09002023408	RUBBER BUFFER	1	

Attach figure 1

System circuit diagram

